
An Energy-Based JEPA World Model of Drug Perturbations

Predicting single-cell state in latent space on Tahoe-100M

Team Dynamics · HackTheWorld(s) · EB-JEPA

The premise — JEPA for cell biology

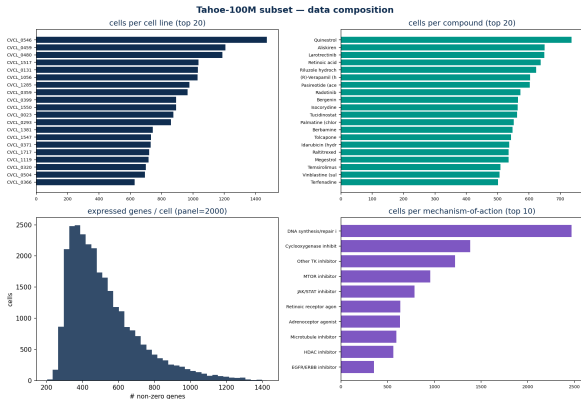
Predict, don't reconstruct

Single-cell transcriptomes are **noisy, sparse, high-dimensional sets** of genes. Reconstructing counts (scGPT) is wasteful; **predicting in a learned latent space** (LeCun's JEPA) discards task-irrelevant noise. *GeneJEPA* (Litman 2025) proved this on Tahoe-100M.

Our goal

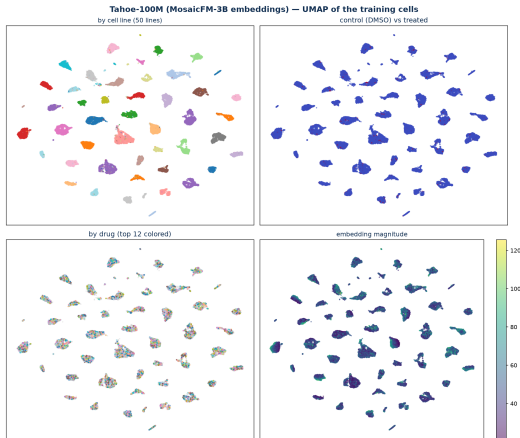
A **true Energy-Based JEPA world model** of drug response — **(control cell, drug) → perturbed cell** — on a **frozen pretrained encoder**, with **biology-specific losses**, beating the standard baselines.

The data — Tahoe-100M: 100M cells, 1,000 lines, 3,000 drugs



Each cell = MosaicFM-3B embedding (2560-d) + drug + cell line + MoA. Preprocessing: top-K gene panel, per-feature z-score, co-expression modules.

The data — UMAP of the cell representations



40k cells: clean structure by **cell line**, a visible **control vs treated** split, drug-level grouping — a healthy space to build a world model on.

The EB-JEPA recipe in one equation

Energy = prediction error in representation space

$$\mathcal{L} = \| g_{\phi}(f_{\theta}(x), q_{\omega}(a)) - f_{\theta}(x') \|^2 + \lambda \mathcal{R}(z)$$

- f_{θ} encoder g_{ϕ} predictor q_{ω} action encoder $\mathcal{R}$ anti-collapse regularizer
- Three settings: image (identity g_{ϕ}) · video (predict next) · **action-conditioned world model** — ours

Our model — frozen foundation encoder + learned drug predictor

- f_θ = **frozen** MosaicFM-3B embedding (2560-d), a pretrained single-cell foundation encoder
- g_ϕ = GRU predictor, **conditioned on the drug** (action a)
- Energy = $\|g_\phi(z_{\text{ctrl}}, a) - z_{\text{pert}}\|^2$ via `JEPA.unroll`
- **Only the predictor is trained**; a drug is a large, controlled state change $\Rightarrow$ **no collapse**

Pipeline

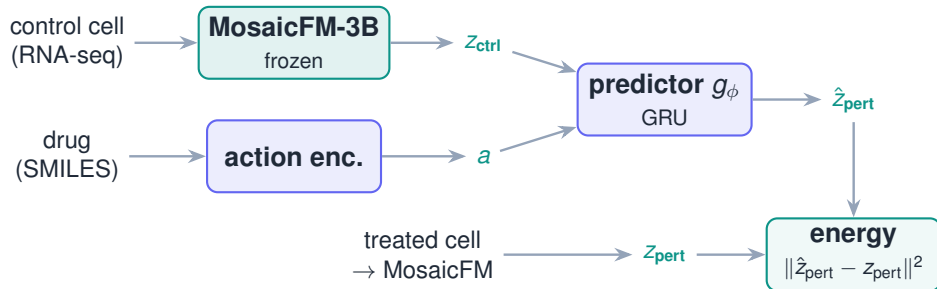
z_{ctrl} (DMSO cell)

$\downarrow g_\phi(\cdot, \text{drug})$

$\hat{z}_{\text{pert}}$

$\downarrow$ energy vs real z_{pert}

Architecture — schematic



Frozen foundation encoder (teal) + **trained** action-conditioned predictor (indigo). Energy = prediction error against the real treated-cell embedding.

Our contribution — three biology-aware loss terms

Track A (trainable encoder): SIGReg + Pathway · Track B (frozen encoder $\Rightarrow$ no anti-collapse):
PerturbationSignature + Pathway

1. SIGReg — anti-collapse (track A; our choice vs GeneJEPA's VICReg)

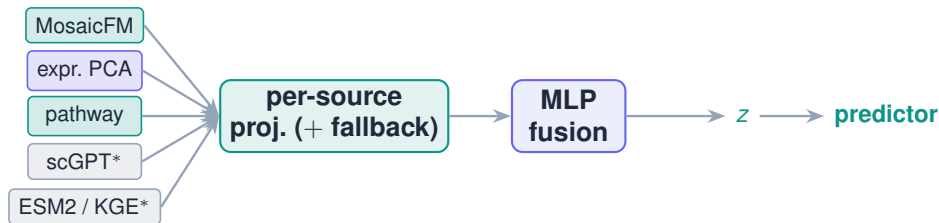
Forces the latent batch toward an **isotropic Gaussian** (Epps–Pulley statistic on random 1-D projections). One hyper-parameter, linear cost, steadier than VICReg's two coefficients.

2. PathwayCoherenceLoss — gene-program prior

Latent pairwise distances must match **co-expression-module activity** distances — injects biological program structure.

3. PerturbationSignatureLoss — drug consistency (track B)

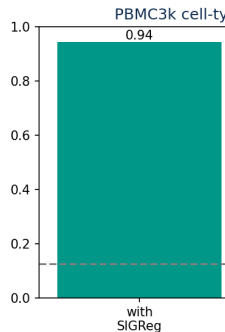
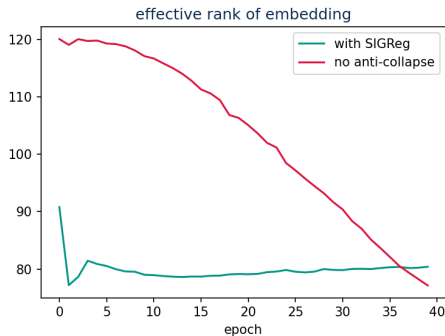
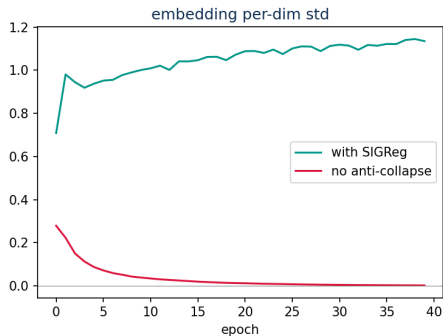
Extension — fuse multiple embedding sources



Each source is **projected** to a common dim, concatenated, fused by an MLP. A **learned fallback** replaces any **missing source** per cell/gene (robust to gaps). * plug-in sources (foundation/sequence/knowledge-graph) — the module accepts any subset.

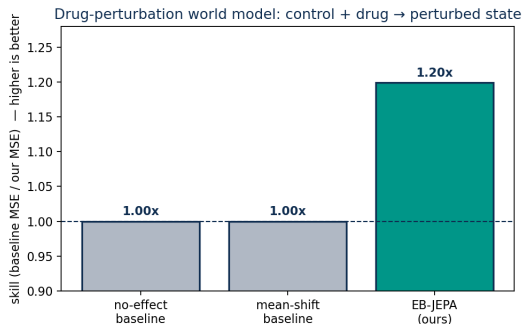
The collapse we fought — SIGReg keeps the encoder alive

Representation collapse: SIGReg keeps the encoder alive



Same cell JEPA on PBMC3k, **with vs without** anti-collapse. Without it: embedding **std** $\rightarrow$ 0.002, effective rank $\rightarrow$ 1, probe $0.43 \approx$ **chance**. With SIGReg: **std** $\approx$ 1.1, **probe** 0.94.

Result — the world model beats both baselines

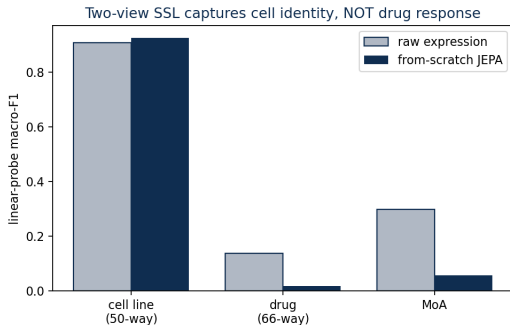


234k train / 59k val cells, real DMSO controls.

predict z_{pert} from	rel. MSE $\downarrow$
no-effect (z_{ctrl})	1.00
mean drug shift	1.00
EB-JEPA (ours)	0.83

skill 1.20 $\times$ vs no-effect, **1.19 $\times$** vs mean-shift. Beating mean-shift = **context-specific** drug response.

Why a world model? Plain SSL learns identity, not drugs



From-scratch two-view JEPA on raw genes (700k cells): **F1 0.93 on cell line** but **0.02 on drug**.

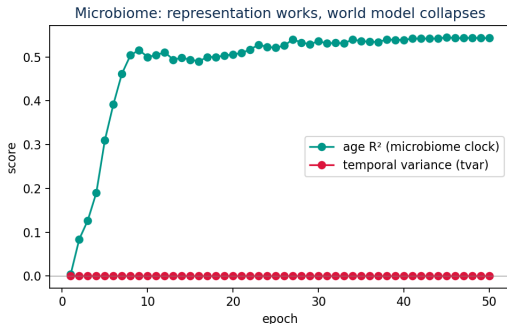
⇒ SSL organizes by the **dominant variance (cell identity)**; **drug response needs the action-conditioned predictor**.

Benchmark — PBMC3k cell type (literature dataset)

method	setting	macro-F1
raw expression	in-domain	0.94
PCA-50	in-domain	0.96
our cell JEPA	in-domain	0.92
GeneJEPA	frozen transfer	0.69
scGPT	frozen transfer	0.23

Honest read: in-domain, raw expression is near-perfect (cell types are linearly separable) — this benchmark doesn't discriminate methods in-domain. GeneJEPA's 0.69 is the harder *frozen-transfer* regime. **Our edge is the perturbation world model, where no trivial baseline exists.**

An honest negative — microbiome dynamics collapse



Same recipe on DIABIMMUNE gut trajectories: representation works (**age** $R^2 \approx 0.53$) but the world model **collapses temporally** ($\text{tvar} \rightarrow 0$, $\text{skill} < 1$).

Insight: VICReg keeps *batch* variance, not *temporal* variance — slow drift makes “no change” unbeatable. Controlled perturbations (Tahoe) remove the problem.

Takeaways

- A **true EB-JEPA world model** of drug perturbation (frozen MosaicFM + learned action predictor) that **beats no-effect & mean-shift baselines (1.2×)** on Tahoe-100M.
- **Three biology-aware design choices**: SIGReg, PathwayCoherenceLoss, PerturbationSignatureLoss.
- A **controlled, honest comparison** + a clean **collapse finding** that motivates the perturbation setting.

Predict cellular response to drugs in latent space — toward in-silico screening.